Reg. No. | | | | | | | | | | | [17CE43]

FOURTH SEMESTER B.TECH EXAMINATION, JUNE–2019
APPLIED HYDRAULIC ENGINEERING

Time: 3 Hours **Maximum Marks: 70**

Instructions:

1. Answer any **FIVE** full Questions, choosing at-least **TWO** from each part.
2. Missing data, if any may suitably be assumed.

PART - A

1. a. What do you understand by the term repeating variables? How are the repeating variables selected for dimensional analysis? **07**

b. A solid sphere of diameter 100mm moves in water at 5.1m/s. It experiences a drag of magnitude 20N. What would be the velocity of 5m diameter spehere moving in air in order to ensure similarity? What will be the drag experienced by it? State which raw governs the similarity given. **07**

γ_{Air} = 1.2kg/m^3, γ_{Air} = 1.86×10^{-5} Pa^{-s}, γ_{water} = 1000kg/m^3, γ_{water} = 1.01×10^{-3} Pa-sec,

2. a. Give atleast 2 reasons why open channel flow is considered more complicated compared to pipe flow? **02**

b. Define the following terms with respect to open channel flow with a neat diagram. **06**

- **i.** Unsteady flow
- **ii.** Non-uniform flow
- **iii.** Rapidly varied flow
- **iv.** Gradually Varied Flow
- **v.** Prismatic channel
- **vi.** Hydraulic radius

c. Calculate average velocity and momentum correction factor (β) for a velocity distribution given by **06**

$u=\left[1+0.5\left(\frac{y}{h}\right)\right]$ for a rectangular channel where u= velocity in m/s atr a distance 'y' from the rectangular channel bed, h= 1m.

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3. a. The section of an open channel consists of a semi circular invert of 600mm radius with vertical tangential sides fig 3(a). If the slope of the channel is 1 in 3000, and depth of water is 1m. Determine the discharge given manning's Roughness coefficient 'n' = 0.012. **07**

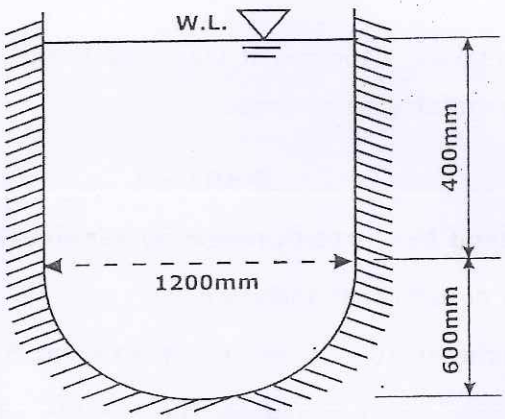

Fig.3(a)

b. Arrange the following materials in descending order of their flow resistance **02**

i. Finesand

ii. Rocks

iii. Pebbles

iv. Fibre reinforced pipe (FRP)

c. How economical sections in open channel is useful for irrigation engineers? Justify your answer for a rectangular section. **05**

4. a. The specific energy for a 5m wide rectangular channel is to 5kgm/kg. If the rate of flow of water through the channel is 25m^3/sec. Determine the alternate depths of channel. **07**

b. Explain classification of surface profiles in open channels with neat sketches. **07**

PART - B

5. a. What is a hydraulic jump? Explain various types of hydraulic jump based on Froud'es No. **07**

b. A gate is suddenly dropped into place closing a rectangular channel 2m deep and 33m wide in which 6m^3/sec of discharger is flowing at a depth of 1.3m. Estimate the depth near the gate and surge velocity. **07**

6. a. Derive an expression of power transmission through pipe. **07**

b. Show the rise of pressure due to sudden closure of a valve in a long pipe line (considering elasticity of pipe material and compressibility of water) is **07**

$$p_i = \frac{V}{\sqrt{\frac{1}{p}\left(\frac{1}{K}+\frac{D}{tE}\right)}}$$ Where

V – AV. Velocity of flow in pipe

F – density of water

D – dia of pipe

K & E – Bulk modules of elasticity of water and young's macula's of elasticity of pipe material

t – thickness of pipe material.

7. a. Write a short notes on any two of the following with reference to CFD. **14**

i. Navier – stokes equation

ii. Grid generation

iii. Applications and limitations of CFD

8. a Distinguish between a pump and turbine can a turbine work like a pump? **04**

b Find the specific speed of turbine installed in a hydro project which develops 20,000kW under a head of 600m, when running at 350 rpm. Specify the type of turbine used. **04**

c. Explain briefly: **06**

i. Draft tube

ii. Priming of a centrifugal pump

iii. Cavitation in hydraulic machines.